Literature Review and Research Positioning

ICT 5143 Capstone, Problem 8: reliable intersectional fairness audits for sparse public-coverage groups

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Version 1.0 · 25 October 2026

Abstract

We review 19 papers on intersectional and subgroup fairness, estimating fairness metrics for small groups, statistical inference for audits, generalisation of fairness constraints, benchmark data and labels, and robust hierarchical shrinkage. The review answers the handout's gap-verification questions. Hierarchical Bayesian estimation of intersectional fairness already exists (Foulds et al. [1]), as does attribute-structured estimation for small subgroups (Herlihy et al. [11]). Contamination and heavy-tailed priors that let outlying units escape shrinkage are classical (Berger and Berliner [19]; Datta and Lahiri [20]). We therefore narrow the contribution to an *evaluation*: when Bayesian pooling produces false "$\Delta\text{EOpp} \leq \epsilon$" conclusions for sparse intersections, including validation-time rule selection. We state it as one falsifiable sentence (H1) and test it out of sample.

1 Search method

We searched arXiv, the ACM Digital Library (FAccT), PMLR (ICML, AISTATS), the NeurIPS and IJCAI proceedings, SIAM (SDM), Project Euclid and journal sites. Search terms combined *intersectional fairness*, *subgroup fairness*, *small groups*, *Bayesian*, *hierarchical*, *shrinkage*, *fairness audit*, *uncertainty*, *contamination prior* and *robust small area estimation*. We started from the references in the handout (Foulds et al., Kearns et al., Ding et al., Folktables) and followed citations. A paper was included if it (i) defines intersectional or subgroup fairness, (ii) estimates or tests fairness metrics under small group sizes, (iii) studies whether fairness guarantees generalise, (iv) supplies our data or label caveats, or (v) supplies the shrinkage method we modify. Every bibliographic entry was checked against its published abstract or proceedings page. For several papers only the abstract was available to us. The table therefore records only what the abstract states and marks other details "not stated in abstract". This is a limitation of the review: rows built from abstracts may omit assumptions stated only in the full text. The closest papers (§4) are the ones whose full text matters most.

2 Thematic review

2.1 Intersectional and subgroup fairness definitions

Kearns et al. [3] show that fairness over a few pre-defined groups can hide unfairness to structured subgroups ("fairness gerrymandering"). They prove that auditing subgroup fairness is computationally equivalent to weak agnostic learning, and they give a learner–auditor game. Hébert-Johnson et al. [4] require calibration on every computationally identifiable subpopulation (multicalibration). Foulds et al. [2] propose *differential fairness*, an intersectional definition motivated by differential privacy. Molina and Loiseau [5] characterise when marginal fairness bounds intersectional fairness, and report that the bounds are loose in practice. Wang et al. [6] discuss which identities to include, how to handle shrinking subgroups, and how to evaluate more subgroups, with experiments on US Census tabular data. Gohar and Cheng [7] survey intersectional notions and mitigation. *Relevance:* our Experiment 1 (single attribute vs intersectional) and our interaction-only semi-synthetic world are instances of the phenomenon these papers establish. They are not new.

2.2 Estimating fairness metrics for small groups

Foulds et al. [1], the reference the handout names as establishing the gap, show that intersectional fairness measurement becomes unreliable as data become sparse. They propose Bayesian models with different bias–variance

trade-offs, including a new hierarchical Bayesian logistic regression, and Bayesian model averaging, evaluated on criminal justice, census and synthetic data. Ji et al. [8] estimate group fairness metrics with a hierarchical Bayesian calibration model that also uses unlabeled data, and report lower estimation error. Morina et al. [9] extend single-attribute metrics to intersections and describe estimation when subgroups have little data. Lum et al. [10] show that naive between-group disparity estimates are statistically biased and propose a "double-corrected" variance estimator. After correction, apparent disparities were no longer significant. Herlihy et al. [11] use structured regression to obtain reliable performance estimates and confidence intervals for very small intersectional subgroups, with goodness-of-fit tests that reveal the structure of harms. Himmelreich et al. [12] argue that the intersectionality problem is one of statistical uncertainty due to small data, and sketch a hypothesis-testing proposal. *Relevance:* uncertainty-aware intersectional estimation, Bayesian or structured, is established. The upward bias of raw max−min gaps that we see in simulation agrees with Lum et al.

2.3 Statistical inference for fairness audits

Cherian and Candès [13] cast auditing as multiple hypothesis testing and use the bootstrap to bound performance gaps over many subgroups at once, with guarantees, either flagging or certifying subgroups. *Relevance:* this is the frequentist counterpart of certifying "$\Delta \leq \epsilon$". We do not reproduce it (see limitations).

2.4 Fairness criteria and generalisation of fairness guarantees

Hardt et al. [14] define equality of opportunity and give post-processing threshold adjustment. Our $\lambda = 1$ rule is the ordinary opportunity-constrained baseline. Cotter et al. [15] show that data-dependent fairness constraints satisfied during training need not hold at evaluation, and they enforce constraints on a separate validation set. *Relevance:* the general failure of fairness constraints to transfer is known. Our narrower question is whether *pooled Bayesian audit criteria* make validation-time tolerance claims that fail on independent test data, and whether they do so more often than raw or independent criteria.

2.5 Benchmark data and label validity

Ding et al. [16] build the Folktables prediction tasks from US Census data across years and states. We use the ACSPublicCoverage task through the Folktables package [17]. Fogliato et al. [18] show that small group-dependent label biases can overturn fairness conclusions. *Relevance:* our label is *absence of public coverage*, not uninsurance; label noise is listed as an untested assumption (D2 in the register).

2.6 Robust hierarchical shrinkage

Berger and Berliner [19] study ϵ-contaminated prior classes and type-II maximum-likelihood (empirical-Bayes) selection of a prior from the class. Datta and Lahiri [20] note that normal-prior Bayes estimators perform poorly with even one extreme small area, and use scale mixtures of normals (e.g. Cauchy for outlying areas) for robust hierarchical Bayes small-area estimation. *Relevance:* letting an atypical group escape shrinkage through a contamination or heavy-tailed prior is classical. **Our contamination prior is an application of this idea, not a new method.**

3 Comparison table

Table 1 compares the papers on the six dimensions required by the handout.

Table 1: Comparison of related work. "n.s." = not stated in the abstract we verified.

Paper	Key assumptions	Data	Graphical / statistical model	Fairness definition	Inference	Limitation for our question
Foulds et al. 2020 (SDM) [1]	Sparse intersections can borrow strength through a probabilistic model	Criminal justice, census, synthetic	Four Bayesian models incl. hierarchical Bayesian logistic regression; model averaging	Intersectional fairness (differential fairness [2])	Bayesian posterior; n.s. beyond that	Outcome-rate fairness, not TPR-based EOpp; masking of an exceptional group and validation-time selection not studied (n.s.)
Foulds et al. 2020 (ICDE) [2]	Protected attributes known; a privacy-style ratio bound is a meaningful fairness notion	COMPAS, census	–	Differential fairness (ϵ-DF)	Learning algorithm satisfying DF	Definition paper; no small-group uncertainty study
Kearns et al. 2018 [3]	Subgroups from a structured function class	n.s. (primarily theory)	–	Subgroup fairness over rich classes	Learner–auditor zero-sum game; agnostic-learning oracle	Auditing is hard in the worst case; finite-sample small-group estimation not the focus
Hébert-Johnson et al. 2018 [4]	Subpopulations computationally identifiable	n.s. (theory)	–	Multicalibration	Algorithms with complexity analysis	Calibration criterion, not EOpp tolerance audits
Molina & Loiseau 2022 [5]	Relations between marginal and intersectional fairness	n.s.	–	Intersectional vs marginal fairness	High-probability bounds; grouping heuristic	Bounds loose in practice (authors)
Wang et al. 2022 [6]	Including more identities creates small subgroups	US Census tabular	–	New intersectional evaluation metrics	Empirical	Pipeline guidance; no Bayesian audit calibration
Gohar & Cheng 2023 [7]	– (survey)	–	–	Taxonomy of intersectional notions	–	Survey
Ji et al. 2020 [8]	Many unlabeled, few labeled examples; scores calibratable per group	Several fairness datasets	Hierarchical latent-variable calibration model	Several group metrics	Bayesian posterior	Groups, not sparse intersections; no tolerance-decision error rates
Morina et al. 2019 [9]	Single-attribute metrics extend to intersections	One real dataset	n.s.	Intersectional extensions of standard metrics	Estimation for small subgroups (n.s.); post-processing	No pooling-masking analysis
Lum et al. 2022 [10]	Group metrics carry sampling noise	Simulation and one real dataset	Variance decomposition (frequentist)	Variance of performance across groups	Double-corrected variance estimator	Summary of spread; not max−min tolerance certification
Herlihy et al. 2024 [11]	Subgroup performance has regression structure in group features	Two public datasets, semi-synthetic variants	Structured regression	Subgroup performance metrics	Confidence intervals; goodness-of-fit tests	Closest structured method; false-tolerance rates under an isolated interaction n.s.
Himmelreich et al. 2025 [12]	Intersectional problem = statistical uncertainty	Partly empirical	–	Desiderata for intersectional fairness	Simple hypothesis testing (sketch)	Conceptual; no calibrated procedure
Cherian & Candès 2024 [13]	Model-agnostic; bootstrap validity	Benchmark datasets	–	Any metric / group criterion	Simultaneous bootstrap bounds (multiple testing)	Frequentist; does not compare Bayesian pooling

Paper	Key assumptions	Data	Graphical / statistical model	Fairness definition	Inference	Limitation for our question
Hardt et al. 2016 [14]	Group-conditional rates estimable	n.s.	–	Equal opportunity; equalised odds	Post-processing threshold adjustment	Small-group estimation error not addressed
Cotter et al. 2019 [15]	Constraints can be enforced on held-out data	n.s.	–	Data-dependent fairness constraints	Two-player constrained optimisation	Training-time generalisation, not audit claims
Ding et al. 2021 [16]	Census-derived tasks represent the population	ACS PUMS, all states, multiple years	–	(benchmark)	–	Dataset; ours uses its ACSPublicCoverage task
Fogliato et al. 2020 [18]	Observed label is a noisy proxy	Two criminal justice datasets	Sensitivity-analysis framework	Predictive bias	Sensitivity analysis	Different domain; motivates label caveat
Berger & Berliner 1986 [19]	Base prior plus ϵ-contamination class	– (theory)	ϵ-contaminated priors	–	Posterior ranges; type-II ML	Not about fairness
Datta & Lahiri 1995 [20]	Some small areas are outliers	n.s.	Hierarchical Bayes with scale-mixture-of-normal priors	–	Robust hierarchical Bayes	Not about fairness or tolerance claims
This project	Exchangeable or attribute-structured pooling of TPRs; binomial counts	ACS PUMS 2018 CA (core), TX (replication)	Discrete BN (predictor); beta-binomial, hierarchical logistic, contamination audit models	Intersectional EOpp (primary), DP (secondary)	VE / clique-tree; EB, grid full Bayes, quadrature; simulation calibration	EB plug-in under-covers; calibration design-specific; struct not an exact reproduction of [1]

4 Closest existing method and baseline reproduction

The closest methods are the hierarchical Bayesian logistic regression of Foulds et al. [1] and the structured regression of Herlihy et al. [11]. Both let a small intersection borrow strength through its attributes. As the handout allows, we reproduce a *clearly documented restricted version* (`struct`, notebook §6):

$$\operatorname{logit}\theta_g = \beta_0 + \beta_{\text{race}[g]} + \beta_{\text{sex}[g]} + \beta_{\text{dis}[g]} + u_g, \qquad u_g \sim N(0,\tau^2), \qquad k_g \sim \text{Bin}(n_g,\theta_g),$$

applied to true-positive rates. β and τ are fitted by marginal likelihood (adaptive quadrature), and u_g is drawn from its grid posterior. Differences from Foulds et al. as far as we could verify: we model TPRs among positives rather than outcome rates; we use only main effects; we plug in empirical-Bayes hyperparameters; and we did not match their priors, covariates or inference algorithm. Exact reproduction is therefore **not** claimed. An exact reproduction (their code, priors and sampler on our counts) was not attempted and remains future work. The quadrature is checked against brute-force integration (10^{-10}). Under its own generative model, its 90% intervals cover 0.665, against 0.905 for the same sampler with the true hyperparameters (notebook §6.1).

5 Gap verification and narrowing of the claim

Candidate gap (handout)	What the literature already answers, and what remains
Hierarchical Bayesian intersectional estimation	Established [1, 8, 11]. Not claimed.
Small groups make fairness metrics unreliable	Established [1, 10, 12]. Not claimed.
Marginal audits can miss intersectional harm	Established [3, 5]. Reported as a replication on our data (Exp. 1).
A prior that lets atypical groups escape shrinkage	Classical in robust and small-area Bayes [19, 20]. Claimed only as an *application*, evaluated as an audit safeguard. We found no application to fairness audits; this search is not exhaustive.
Fairness guarantees may not generalise from training to test	Established for training constraints [15]. **Remaining:** whether *pooled audit criteria* produce validation-time tolerance claims that fail on test (H1), and how this compares with raw and independent criteria.
Pooling masks a small exceptional group; quantile rules as a fix; calibrating a tolerance procedure	Not answered in the papers we reviewed. We add a diagnosis and a tested fix (design-matched calibration, held-out certification). **Remaining:** a controlled evaluation of false-tolerance rates by deficit structure (cell, interaction-only, main effect), thinning and design, and whether a simulation calibration transfers to the deployment design.

Narrowed contribution. A controlled and real-data evaluation of when Bayesian pooling, exchangeable and attribute-structured, produces false "ΔEOpp $\leq \epsilon$" conclusions for sparse intersections. It covers minority thinning, interaction-only and main-effect deficits, validation-time rule selection, and the transfer of a simulation-calibrated robust-prior procedure between designs.

6 Falsifiable contribution

H1. *When the group-threshold rule's λ is selected on validation data and a pooled Bayesian audit criterion (hierarchical, full-Bayes, complete, attribute-structured or contamination pooling) reports intersectional $\Delta EOpp \leq 0.10$, the frozen rule's empirical test $\Delta EOpp$ exceeds 0.10 in more than half of those cases.*

H1 was formed from the California results and tested out of sample on Texas 2018 (notebook §14.1), with the decision rule fixed before the test was run.

Texas 2018, 5 household splits	value
pooled selections	40
validation-time claims ΔEOpp ≤ 0.10 (pooled)	27
claims failing on test (ΔEOpp > 0.10)	27
test ΔEOpp of the claims	0.110–0.888
claims by raw / independent criteria	0
H1 verdict	**supported**

Methodological contribution and its test (H2). The review found no application of robust priors to fairness audits and no study of calibration transfer between audit designs. We therefore propose *design-matched calibration* plus *held-out certification* (notebook §15) and test it with preregistered hypotheses on the Texas design. Its audit score $J = \mathrm{ACC} - \mathrm{FTS}$ is 0.664, against at most 0.385 for every established audit (H2a supported) and 0.189 for the invented-design calibration (H2b supported). Held-out certification produced no false tolerance claims on real CA or TX data, against 10 of 10 with same-data certification (H2c supported).

Earlier draft claim. The draft claim "an ϵ-contamination prior reduces the false-satisfaction rate of an empirical-Bayes upper-quantile audit by at least half" is not supported for the calibrated procedure ($0.980 \rightarrow 0.503$; the required value is ≤ 0.490).

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